PCB Design & Simulation Challenge

Theme: Op-Amp / 555 Timer Circuit

1. EVENT OVERVIEW

This competition tests your ability to read datasheets, design custom electronic symbols, perform calculations, and validate your designs through simulation.

- Platform: EasyEDA (Standard Edition)
- Team Size: 2
- Time Limit: 3 Hrs
- Date & Time: 18/05/2026

2. GENERAL RULES

1. Software: All design and simulation must be done strictly on EasyEDA.
2. Plagiarism: Participants must build their custom symbols from scratch and schematics from given ref. Using pre-made community symbols for the main IC will result in immediate disqualification.

3. EVENT WORKFLOW & TASK RULE

Upon starting the event, participants will be provided with a Problem Statement Kit containing a reference schematic, an IC datasheet, partial R&C values, and a specific mathematical formula.
You must strictly follow these five phases:

Phase 1: Custom Symbol Design

- Rule 1.1: You cannot use the default EasyEDA library symbol for the main IC (Op-Amp or 555 Timer).
- Rule 1.2: You must create a custom schematic symbol from scratch.
- Rule 1.3: You must create a custom symbol footprint from scratch.
- Rule 1.4: The pin names, numbers, and layout must accurately reflect the official datasheet provided in the kit.

Phase 2: Component Calculation

- Rule 2.1: The reference schematic will feature one unknown Resistor (R) or Capacitor (C) value.
- Rule 2.2: You must calculate the missing value using only the formula and parameters provided.
- Rule 2.3: Calculation steps must be documented clearly. Guesswork or trial-and-error in simulation without mathematical proof will result in a loss of points.

Phase 3: Schematic Capture

- Rule 3.1: Draft the exact reference circuit in EasyEDA.
- Rule 3.2: You must insert and use your custom-designed IC symbol.
- Rule 3.3: Use the calculated R/C value from Phase 2. Standard EasyEDA libraries can be used for basic passives (resistors, capacitors, power sources, grounds).
- Rule 3.4: Ensure the schematic is neatly wired, avoiding overlapping lines and floating pins.

Phase 4: PCB Layout Creation

- Rule 4.1: Once the schematic is simulated and verified, convert the schematic to a PCB layout.
- Rule 4.2: You must link an appropriate physical footprint (e.g., DIP-8, SOIC-8) to your custom IC symbol, as defined by the provided datasheet.
- Rule 4.3: **No auto routing is allowed.** All copper tracks must be routed manually. Use of the auto-router tool will result in disqualification.
- Rule 4.4: Component placement should be logical and compact. Ensure proper track widths are used (e.g., thicker tracks for power/ground, standard for signals).
- Rule 4.5: The final board must pass the Design Rule Check (DRC) with zero errors (no unrouted nets, clearance violations, or overlapping tracks) and area of board.

Phase 5: Circuit Simulation

- Rule 5.1: Setup the appropriate simulation environment (Transient/DC/AC) as required by the circuit.
- Rule 5.2: Attach voltage/current probes at the appropriate input and output nodes.
- Rule 5.3: Run the simulation to generate a waveform that proves the circuit functions correctly based on your calculated values.

4. GROUNDS FOR DISQUALIFICATION

- Using automated symbol generators or pre-existing EasyEDA library symbols for the target IC.
- Submitting broken project links or private projects that judges cannot access.
- Any form of cheating, sharing project links with other teams, or submitting after the deadline.

5. Learning Resources or EasyEDA:

Description	Youtube Link
EasyEDA Playlist 001 - 014	https://www.youtube.com/watch?v=MsuR6W-jN5M&list=PLbKMtvtYbdPMZfzGuVTdc0MWKrFvU4nsu&index=1
EasyEDA Full TUTORIAL + Create Component + TIPS	https://www.youtube.com/watch?v=utBQqcuOt9U
How To Use EasyEDA, A Free Circuit Simulator	https://www.youtube.com/watch?v=9gdtav5fx88
Build Your First Inverting Amplifier Schematic (Step-by-Step)	https://www.youtube.com/watch?v=8ukjX9TK9-s
How To Create a 555 flasher based PCB	https://www.youtube.com/watch?v=rG0RtIj4siU